BookFlix

Project Documentation

Web App Development

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A. Technologies Used

BookFlix has three parts: a React frontend, Firebase for accounts and data, and a Node.js server for the AI features.

Frontend

- React 19 (Create React App): the user interface, built from components, with state handled through hooks (useState, useEffect, useContext).
- React Router 7: navigation between pages, routes with parameters (/item/:id) and protected routes.
- React Context: global state for the account, cart, favorites, library and admin role.
- React.lazy and Suspense: heavier pages are downloaded only when they are opened.
- CSS with custom properties: colors, fonts and a responsive layout using Grid, Flexbox and media queries.
- Google Fonts: Playfair Display for headings and Inter for text.

Cloud (Firebase)

- Firebase Authentication: accounts with email and password, Google sign-in and password reset.
- Cloud Firestore: NoSQL database for books, users, reviews and the community feed, with real-time updates.
- Firestore Security Rules: server-side access control for every read and write.

Backend

- Node.js 24 and Express 5: the REST API for the AI features, including streamed responses.
- Google Gemini (@google/genai): the language model behind the story maps, character chat, Plot Detective and the assistant.
- firebase-admin: checks the sign-in token sent by the browser.
- Zod: validates incoming requests and the JSON answers from the AI.

Other tools and data

- Firebase CLI: used to publish the security rules.
- Open Library Covers: source of the book cover images.

B. Implemented Features

1. Home Page

- Welcome section with a looping video background, the BookFlix title and a short tagline.
- Search bar: filters the catalog live, as you type, by title, author or director.
- “Our collection” heading with the number of books currently shown.
- Sort menu: Featured, Title A–Z, Price low to high, Price high to low, Newest first, Oldest first.
- Genre filter: chips for 9 genres (Classics, Mystery & Thriller, Fantasy, Literary Fiction, Young Adult, Romance, Sci-Fi & Dystopia, Historical Fiction, Non-fiction), each with its number of books. It combines with the search.
- Book cards in a responsive grid, each with:
  - the cover and a badge with the main genre;
  - a heart button to add or remove the book from favorites;
  - title, author, year, and the price (or “In your library” if the reader already owns it).
- Loading skeletons while the books are fetched from Firestore.
- Community feed (“From the reading community”): the latest 12 progress updates shared by readers, updated in real time, with links to the books and how long ago they were posted.

2. Navigation Bar

- Sticky bar at the top of every page with the BookFlix logo (link to Home).
- Links: Home, By Vibe, My List, Favorites (with the number of favorites), Cart (with a badge for the number of items).
- Admin link, shown only to administrator accounts.
- When signed in: the user's initial or Google photo and name, linking to the account page, plus a Logout button. When signed out: a Sign in link.
- The current page is highlighted.
- On phones the links fold behind a menu button, which also shows the cart count.

3. Sign In and Create Account

- Two-column card: a library photo with a quote on the left, the form on the right (one column on phones).
- Two tabs: Sign in and Create account.
- Continue with Google button (Firebase Google sign-in).
- Email and password form; account creation also asks for a name.
- Show/Hide button for the password; minimum 6 characters for new accounts.
- Forgot password? sends a reset link to the email typed in the form.
- Clear error messages instead of Firebase codes (for example “That email and password don't match”).
- The user's profile is saved in Firestore (users/{uid}).
- After signing in, the user returns to the page they came from (for example the checkout), or goes to their account.

4. Book Page

- Back link, large cover, genre tags, title, author and year.
- Average rating in stars, with the number of reviews.
- Description of the book.
- Buying options:
  - Buy the book (54.99–89.99 RON): adds it to the cart;
  - Read online (about a third of the price) for 30 days of access.
- If the reader already owns or rents the book, a banner shows it, with a Read now button.
- Add to favorites, + My List, and a shortcut to the cart.
- Story features: links to Enter the Book, Character Chat and, for mysteries, Plot Detective.
- Reading progress: a chapter picker (“I've finished Chapter 5: …”) with previous/next buttons and a progress track; it saves automatically. Titles of chapters not yet reached stay hidden to avoid spoilers.
- Share your progress: posts a sentence such as “Cristina finished chapter 5 of 19 in The Hobbit” to the community feed, with an optional note.
- Reviews:
  - signed-in readers give 1 to 5 stars and write a review; one review per reader, which they can update;
  - each review shows the author's photo or initial, name, date and stars; the reader's own review is marked “(you)”.

5. Book by Vibe

- Find books by atmosphere instead of genre.
- Quick moods: Rainy night mystery, Blanket & tea, Grand adventure, Can't-sleep thriller.
- Three groups of vibes to combine: Setting & weather (rain, city at night, small town…), Mood (mystery, cozy, dark, hopeful…), Themes (coming of age, friendship, survival…).
- A summary of the selected vibes, with a Clear button.
- Results ranked by match: each book shows a "% match" badge and the vibes it matched.

6. Cart

- Each line shows the cover, title, author, the type (Book purchase or Read online · 30 days), the price and a Remove button.
- Order summary: subtotal, delivery ("Instant · Free") and total.
- Buttons: Proceed to checkout and Continue browsing.
- An empty cart shows a message with a link to the collection.
- The cart is saved on the user's account, or in the browser for guests; a guest's cart moves into their account at sign-in.

7. Checkout

- Requires an account: guests see "Sign in to finish your order" and come back after signing in.
- Your details: full name and email, prefilled from the account.
- Payment method: Card, PayPal or Apple Pay. Payment is simulated; no card details are collected.
- Order summary with each item and the total, and a "Pay … with …" button.
- Confirmation page: "Thank you", the order total and a Start reading button for every book bought.
- Purchased books go to the user's library; online access gets an end date 30 days later.

8. Online Reader

- Opens only for readers who own or rent the book; anyone else sees a locked page with a link to the buying options.
- Book-style layout with the title, author and the text of the first chapter (the book's content, or its description).
- A− / A+ buttons to change the text size.
- Day / Night mode.

9. Favorites

- Grid of every book the reader marked with the heart, using the same cards as the home page.
- The heart can be toggled from any book card or book page.
- Empty state with a link to the collection.

10. My List

- A personal reading list (“books you're planning to read next”), available only when signed in.
- Books are added with “+ My List” on the book page; a book can't be added twice.
- Each entry shows the cover, title, author and year, with a Remove button.

11. Account Page

- Available only when signed in.
- Profile card: photo or initial, name, an Admin badge for administrators, email and “Reader since” date.
- Buttons: Edit name, Change password (sends a reset email; for email accounts) and Log out.
- Statistics: books owned, books read online, favorites, items in the cart; each tile links to its page.
- Continue reading: up to 4 started books, with the chapter reached, a progress bar and a link to the story map.
- My library: purchased books (with the purchase date) and online rentals (with the end date), each with a Read button.
- Favorites: the first four favorites, with a link to all of them.

12. Enter the Book (Story Map)

- An interactive, spoiler-free map of the book's world, generated by the AI the first time the book is opened and then saved for everyone.
- A short premise and the chapter picker: the map shows only what the reader has reached.
- Map on a parchment background with a compass:
  - unlocked places are pins with an icon and a name, joined by a dashed path;
  - places from later chapters are “?” pins showing the chapter they appear in.
- Clicking a place opens a side panel with its description, who you meet here (with a link to talk to them) and what happened here.
- Characters you've met: a list linking to Character Chat.
- Your journey so far: a timeline of the unlocked events, each marked with its chapter.
- A hint shows how many items are still hidden and the chapter where the next one appears.

13. Character Chat

- Conversations with the characters of a book, played by the AI.
- The same chapter picker; only characters the reader has already met are listed, and the number still to come is shown.
- The chat window shows the character's name and role, message bubbles and a typing indicator; replies appear word by word.
- The character keeps their personality and only knows events up to the reader's chapter; questions about later events are deflected in character.
- Characters reply in the language the reader writes in (for example Romanian).

14. Plot Detective

- Available for books tagged as mysteries; the reader plays detective while reading.
- The case: the central question of the book, without the answer.
- The chapter picker unlocks suspects and clues as the reader progresses.
- Evidence board styled as a corkboard:
  - suspect cards (name, role, description, an Accuse button);
  - clue cards (title, text, the chapter where it was found);
  - sticky notes the reader writes;
  - cards can be dragged anywhere; “Connect with string” links two cards with a red string.
- Your theory: who did it (a list of the suspects) and how and why (free text).
- Close the case: the AI compares the theory with the real solution and returns a score from 0 to 100, a verdict, the answer, the full solution, what the reader got right and what they missed. Revealing before the last chapter asks for confirmation, since it spoils the ending.
- The board is saved per user, so it is there when the reader comes back.

15. BookFlix Assistant

- A small round button in the bottom-right corner of every page, which opens a chat panel.
- Suggested questions to start with, for example “Recommend a cozy book for a rainy evening”.
- Recommends only books from the BookFlix collection, with links to their pages, and explains how the site works.
- Knows the current page, the cart and the favorites, so it can answer “What's in my cart?”.
- Replies stream word by word; Start over and Close buttons.
- Available to guests as well as signed-in users.

16. Admin Dashboard

- Page /admin, open only to administrators; others see an “Admins only” message.
- Overview tab:
  - totals for books, readers, reviews, feed posts, sales and revenue;
  - rankings of books per genre, best sellers and top-rated books.
- Books tab (CRUD):
  - a searchable table with cover, title, author, genres, year and price;
  - Add book and Edit open a form with title, author, year, price, cover link with live preview, description, genres and vibes;
  - validation with messages under each field (required fields, valid year, https image link, description of at least 20 characters, at least one genre);
  - an option to rebuild the book's story map and case file after big changes;
  - Delete removes the book and its reviews, after a confirmation.
- Reviews & feed tab: the latest reviews and feed posts, each with a Delete button for moderation.

17. Page Not Found

- Any unknown address shows a styled “This page isn't on our shelves” message with a link back to the collection.

C. Data, Security and the AI Server

Database (Cloud Firestore)

- item/{bookId}: book details, genres, vibes, optional price, and the saved story map and case file.
- item/{bookId}/reviews/{uid}: one review per reader, with rating, text, name, photo and date.
- activityFeed/{postId}: posts in the community feed.
- users/{uid}: the user's profile (name and email).
- users/{uid}/private/shop: cart, favorites, purchased and rented books.
- users/{uid}/reading/{bookId}: reading progress for each book.
- users/{uid}/list/{bookId}: the books in My List.
- admins/{uid}: the administrators, editable only from the Firebase console.

Security

- Firestore Security Rules are published with the Firebase CLI. Anyone can read the catalog, reviews and feed, but only administrators can change books.
- Each user can read and change only their own data. Reviews and feed posts can only be written in the user's own name.
- After publishing, the rules were tested with a signed-out client: it could read the catalog, but could not change books, make itself an administrator, list users, read another user's progress or post to the feed.
- My List and the account page send guests to the sign-in page. Checkout requires an account, and the admin page checks the administrator role.
- The Gemini API key is kept on the server in server/.env, which is excluded from Git.
- Every request to the AI server carries the user's Firebase ID token, which the server checks with firebase-admin.
- The AI server allows at most 40 requests per 10 minutes for each user, to protect the free quota, and validates all data with Zod.

AI Server Endpoints

- GET /api/health (public): tells whether the AI is configured.
- POST /api/world (signed in): generates a book's story map, tagged by chapter.
- POST /api/chat (signed in): streams a character's reply, limited to the reader's chapter.
- POST /api/case (signed in): generates the suspects and clues of a mystery.
- POST /api/case/reveal (signed in): scores the reader's theory and reveals the solution.
- POST /api/assistant (public): streams the site assistant's reply.
- Maps and case files are requested from Gemini as JSON with a fixed structure and are validated before use.
- If the main model is overloaded or out of quota, the server tries again and then switches to a lighter model.

D. Running the Project

- Requirements: Node.js 20 or newer and a free Gemini API key from Google AI Studio.
- Install the dependencies: npm install
- Copy server/.env.example to server/.env and set GEMINI_API_KEY.
- Start the AI server: npm run server
- In a second terminal, start the website: npm start, then open http://localhost:3000.

Payment in the shop is simulated. AI-generated content is cached per book and can be rebuilt from the admin dashboard.